BSAVA MACH Architecture: Power BI Reporting & Data Lake Strategy

1. Executive Summary

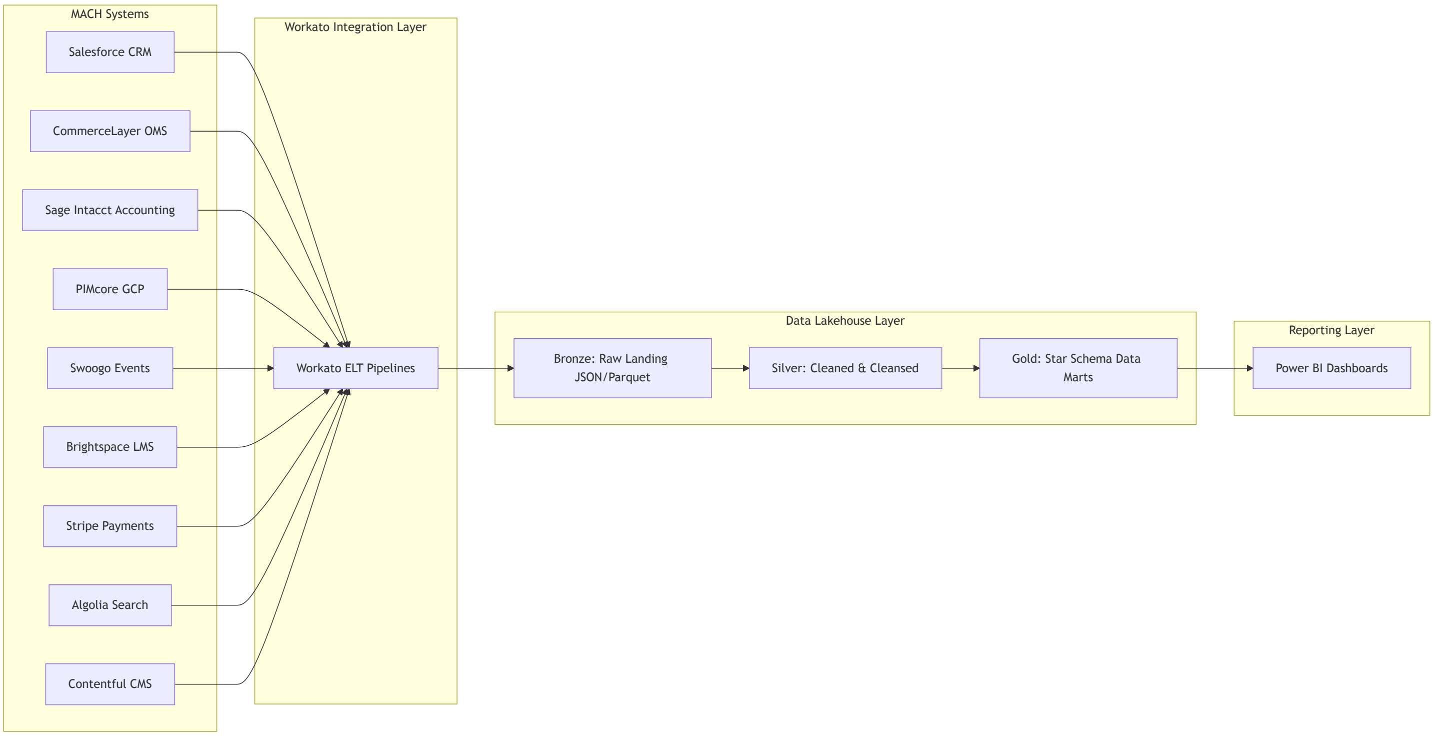
The BSAVA Headless MACH Architecture integrates best-of-breed microservices (**Salesforce**, **CommerceLayer** OMS, **Sage Intacct** Accounting, **PIMcore**, **Contentful**, **Swoogo**, **Brightspace**, **Stripe**, and **Algolia**). While this composable approach delivers agility and performance for the digital frontend, enterprise reporting in **Power BI** requires centralized, aggregated datasets.

This document outlines:

1. **Datasets & Source Mapping**: Key domain datasets (including Order Management and Financial Accounting) to aggregate for business intelligence.
2. **Workato Ingestion Strategy**: Recommended ELT pattern, triggers, and synchronization frequencies across all microservices.
3. **Data Lake Evaluation**: Strategic comparison of **Microsoft Fabric** vs. **Snowflake** for powering Power BI.

2. Dataset Aggregation Requirements for Power BI

To provide unified business metrics—such as Member 360, Order Management Analytics, GAAP Revenue Recognition & Financial Ledger Reconciliation, Event ROI, and Learning Analytics—data must be extracted from the isolated MACH services and modeled into a Star Schema (Dimensional Model) in the central data lake.



2.1 Domain Dataset Matrix

Reporting Domain	Primary Source Systems	Key Entities & Data Fields	Power BI Report Use Cases
Member 360 & Engagement	Salesforce, CommerceLayer, Swoogo, Brightspace, Stripe	Member ID, Contact details, Membership type/tier, Order history, Lifetime Value (LTV), CPD hours, Engagement score	Member retention analysis, churn risk prediction, full lifetime purchase tracking
Order Management & E-Commerce	CommerceLayer (OMS), Stripe, PIMcore, Salesforce	Order IDs, Line Items, SKUs, Unit Prices, Discounts/Promotions, Shipping/Fulfillment status, Cart abandonment, Order states	E-commerce sales volume, Average Order Value (AOV), fulfillment velocity, promo effectiveness, SKU sales breakdown
Financial & Revenue Reconciliation	Sage Intacct, Stripe, CommerceLayer, Salesforce, Swoogo	GL Account Entries, Accounts Receivable (AR), Invoices, Payment Payouts (Stripe), Revenue Recognition	GAAP revenue recognition, GL ledger reconciliation (CommerceLayer/Stripe to Sage Intacct), MRR/ARR,

Reporting Domain	Primary Source Systems	Key Entities & Data Fields	Power BI Report Use Cases
		Schedules, Tax Summaries	financial audit readiness, P&L analysis
Event Analytics & Delegate Tracking	Swoogo, CommerceLayer, Salesforce, PIMcore	Event IDs, Registration timestamps, Ticket types, Attendance check-ins, Revenue per event, Speaker & Session ratings	Event registration trends, member vs non-member attendance, event profit/loss
Education & CPD (LMS)	Brightspace LMS, Salesforce, PIMcore	Course enrollments, Progress %, Completion dates, Assessment scores, Continuing Professional Development (CPD) hours	CPD compliance reporting, course popularity, learner retention & completion rates
Product & Catalog Performance	PIMcore, CommerceLayer, Stripe, Swoogo, Brightspace	Product SKUs, Digital Assets, Bundles, Pricing tiers, Category taxonomies, Cross-domain inventory	Bestselling publications/courses, bundle conversion, inventory performance
Content & Search Intelligence	Contentful, Algolia, Next.js Web Analytics	Page views, Article publishing dates, Search terms, Zero-result searches, Click-Through Rate (CTR), Conversion path	Content engagement, search gap analysis, demand discovery for new courses/events

3. Workato Data Aggregation Strategy

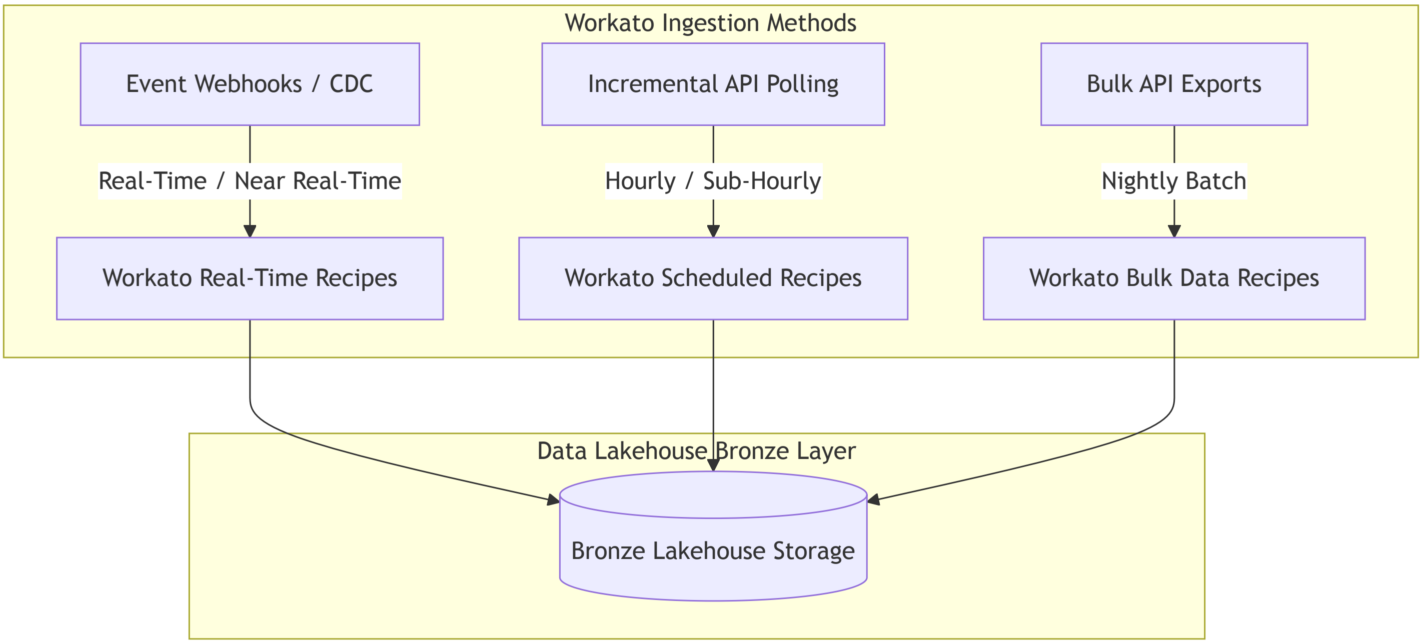
Workato serves as the enterprise integration platform (iPaaS) connecting source APIs to the Data Lakehouse.

3.1 Architecture Pattern: ELT over ETL

It is strongly recommended to adopt an **ELT (Extract, Load, Transform)** pattern rather than transforming data within Workato recipes:

- **Extract & Load (Workato)**: Workato extracts raw data from MACH APIs and loads it directly into the Data Lakehouse **Bronze Layer** (as JSON/Parquet/Delta).
- **Transform (Lakehouse Engine)**: Data cleansing, deduplication, schema enforcement (Silver Layer), and star-schema dimensional modeling (Gold Layer) are executed natively within the Data Lakehouse using **SQL / dbt / Spark**.

3.2 Ingestion Patterns & Frequencies



1. **Real-Time / Event-Driven Webhooks (Bronze Landing)**:
   - **Sources**: CommerceLayer (`orders.create` , `orders.fulfill` , `orders.cancel`), Stripe (`charge.succeeded` , `payout.created`), Swoogo (Registrations), Salesforce (Member Status Updates).
   - **Mechanism**: Source webhooks trigger Workato HTTP endpoints. Workato appends raw JSON payloads directly to Lake Storage.
   - **Latency**: < 1 minute.
2. **Incremental Polling (Scheduled Batch)**:
   - **Sources**: Sage Intacct (GL Journal Entries, AR Invoices, Customer Ledger updates), Brightspace (Progress updates), Algolia (Analytics metrics), PIMcore (Catalog changes).
   - **Mechanism**: Workato recipes execute hourly using high-watermark timestamps (`updated_at >= last_sync_time`).
   - **Latency**: 1 hour.
3. **Nightly Bulk Syncs**:

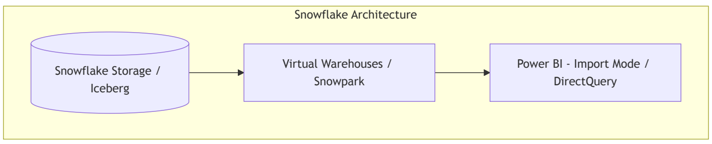
- **Sources**: Sage Intacct (Full financial period ledgers & trial balances), Salesforce Bulk API 2.0 (Full object snapshots), Brightspace Data Hub (CSV export dumps).
- **Mechanism**: Scheduled nightly Workato bulk recipes stream high-volume CSV/Parquet dumps into the lake.
- **Latency**: 24 hours.

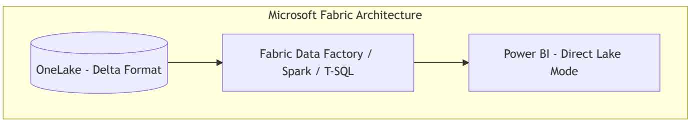
3.3 Workato Recipe Best Practices

- **Idempotency & Replayability**: Always include source record IDs (e.g. CommerceLayer Order ID, Sage Intacct Document Number) and ingestion timestamps. Ensure lake ingestion operations are append-only or upsert-safe.
- **Payload Preservation**: Store the un-truncated JSON/Avro payload in the Bronze layer to safeguard against source API schema changes.
- **Rate Limit & Error Handling**: Implement exponential backoff, circuit breakers, and dead-letter queues (DLQ) in Workato for source API rate limits (e.g. Salesforce API governors, Sage Intacct XML Web Services concurrency limits).

4. Data Lake Candidate Evaluation: Microsoft Fabric vs. Snowflake

Both **Microsoft Fabric** and **Snowflake** are enterprise-grade platforms capable of acting as the central data lakehouse for BSAVA. However, they possess distinct architectural philosophies and Power BI integration characteristics.





4.1 Detailed Architectural Comparison

Dimension	Microsoft Fabric	Snowflake
Primary Paradigm	Unified SaaS Lakehouse (OneLake / Delta Lake)	Multi-Cloud Data Cloud / Data Lakehouse
Power BI Integration	**Direct Lake Mode** (Queries Delta tables directly in RAM without importing or translation)	**Import Mode** (Scheduled refresh) or **DirectQuery** (SQL queries translated per visual interaction)

Dimension	Microsoft Fabric	Snowflake
Query Latency in Power BI	**Sub-second** (Direct Lake matches Import speed without refresh delays)	Fast in Import mode; variable in DirectQuery (dependent on virtual warehouse sizing)
Data Format	Open format by default (**Delta Lake / Parquet**)	Proprietary micro-partition format or Apache Iceberg tables
Workato Connector Support	Supported via OneLake REST API, ADLS Gen2 connector, or Fabric SQL Endpoint	Highly mature native Snowflake connector with `COPY INTO` bulk staging support
Transformations / Modeling	Data Factory Pipelines, Notebooks (PySpark/Scala), SQL Analytics Endpoint, dbt-fabric	SQL, Snowpark (Python, Java, Scala), Dynamic Tables, dbt-snowflake
Security & Governance	Microsoft Purview, Microsoft Entra ID (Azure AD), RLS/CLS native across Fabric & Power BI	Snowflake Horizon, Native RLS/CLS, Role-Based Access Control (RBAC)
Licensing & Pricing	Capacity-based (F-SKUs) sharing pooled compute across ETL, Spark, Lakehouse, and Power BI	Usage-based credits (Compute Virtual Warehouses) + Storage billed separately per GB

4.2 Deep-Dive: Microsoft Fabric Pros & Cons

Advantages of Microsoft Fabric for BSAVA

1. **Direct Lake Mode for Power BI**:
   - Power BI reports read Delta Parquet files directly from OneLake without requiring data duplication or scheduled dataset refreshes.
   - Eliminates 50 GB Power BI dataset size limits and long refresh windows.
2. **Unified Data & Analytics Ecosystem**:
   - Combines Data Engineering (Spark), Data Integration (Data Factory), Data Warehousing, and BI (Power BI) under a single tenant and capacity subscription.
3. **Native Microsoft Ecosystem Synergy**:
   - Seamless authentication with Microsoft Entra ID, security compliance with Purview, and native connectivity to Power Platform / Microsoft 365.
4. **Single Source of Truth (OneLake)**:

- Eliminates data silos; all workspaces share the underlying OneLake storage in open Delta format.

Disadvantages of Microsoft Fabric

1. **Platform Maturity**:
   - Fabric is newer compared to Snowflake; certain niche administrative features and dbt adapters are actively maturing.
2. **Vendor Lock-in to Microsoft Ecosystem**:
   - Highly optimized for organizations heavily invested in Microsoft enterprise tooling.

4.3 Deep-Dive: Snowflake Pros & Cons

Advantages of Snowflake for BSAVA

1. **Industry-Proven Stability & Maturity**:
   - Exceptionally robust performance, zero-copy cloning, time travel, and seamless multi-cloud deployment (AWS, GCP, Azure).
2. **Mature Workato Ecosystem**:
   - The Workato Snowflake connector is among the most battle-tested in the industry, supporting automated staging (e.g. S3/GCS) and high-throughput bulk loading.
3. **Snowpark & Dynamic Tables**:
   - Advanced Python/Java data processing pipelines natively inside the database, simplify continuous ELT updates via Dynamic Tables.
4. **Cloud Agnostic Architecture**:
   - Runs natively on GCP (where BSAVA's PIMcore resides) or AWS/Azure, avoiding reliance on a single cloud vendor's stack.

Disadvantages of Snowflake

1. **Power BI Overhead & Trade-offs**:
   - Power BI cannot query Snowflake directly without either using **Import Mode** (which requires dataset refresh schedules and storage duplication) or **DirectQuery Mode** (which keeps virtual warehouses running, increasing credit costs).
2. **Dual Licensing / Cost Complexity**:
   - Separate billings and management for Snowflake compute credits + Power BI Premium/Fabric capacities.

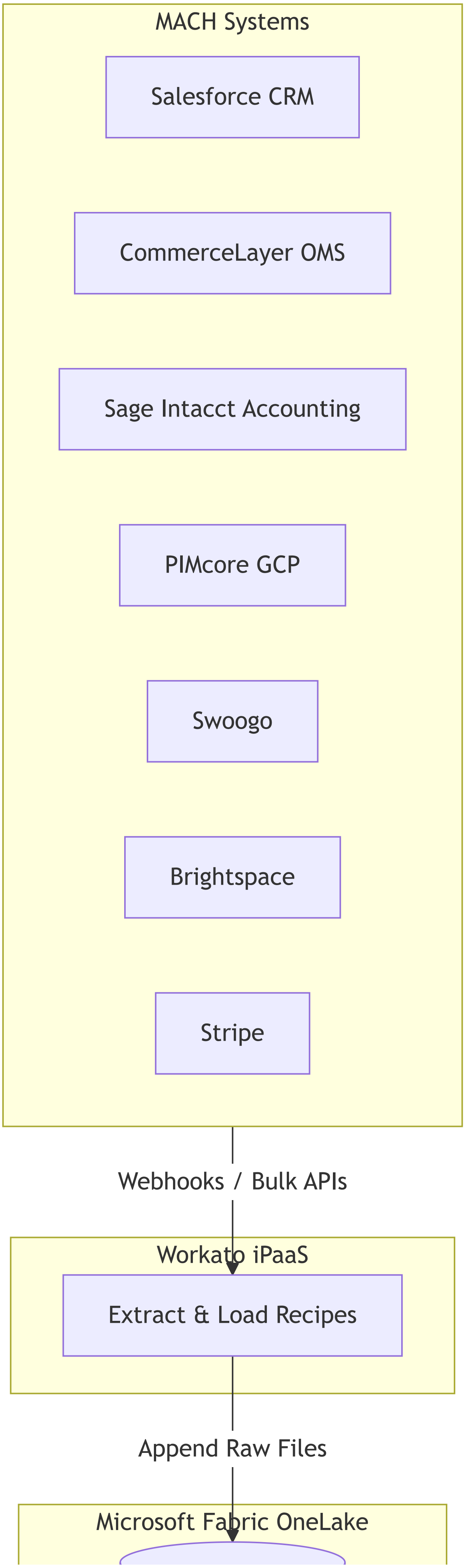
5. Strategic Recommendation & Architecture Decision Matrix

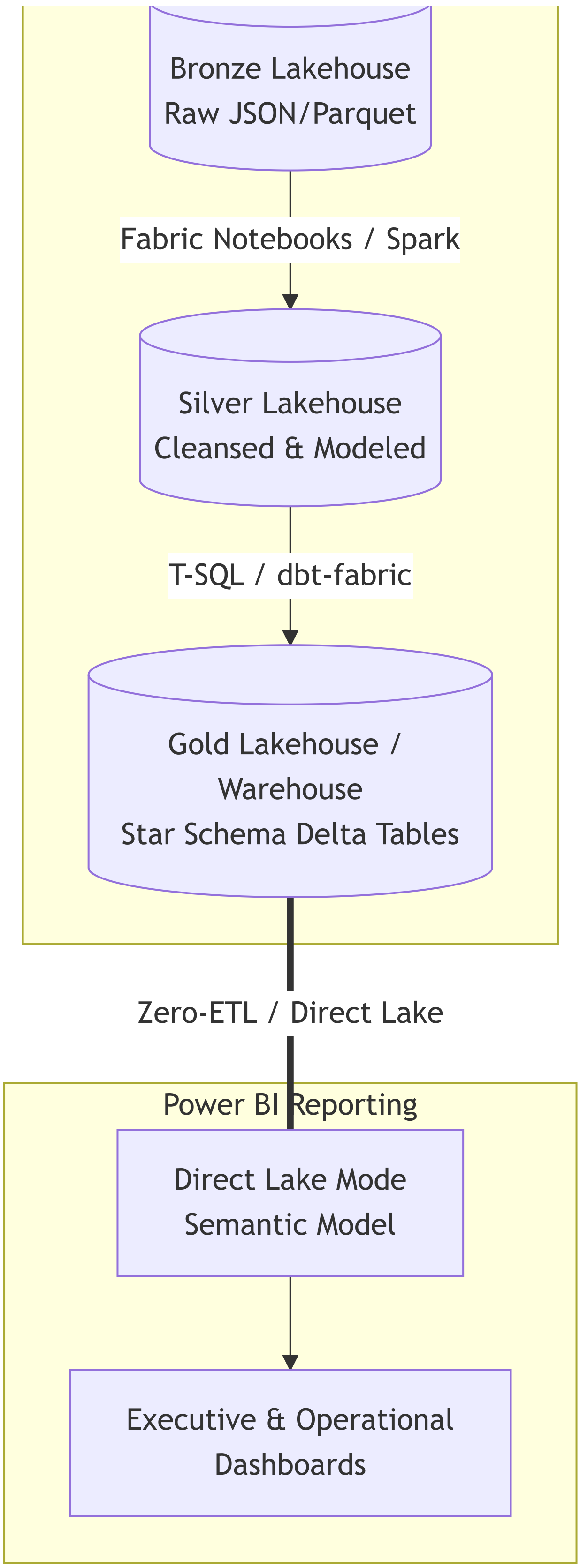
Decision Recommendation: Microsoft Fabric

For BSAVA's reporting requirements with **Power BI as the designated BI front-end**, **Microsoft Fabric is the recommended choice**.

Rationale:

1. **Unmatched Power BI Performance (Direct Lake)**: Because Power BI is the user's explicit reporting tool, Fabric's **Direct Lake mode** provides real-time access to enterprise-scale data in OneLake with native memory performance, eliminating data model refreshes and DirectQuery credit overhead.
2. **Total Cost of Ownership (TCO)**: Fabric consolidates data warehousing, lake storage, and BI capacity into a single predictable Microsoft F-SKU capacity, avoiding dual billings for Snowflake virtual warehouses + Power BI Premium.
3. **Workato Integration Feasibility**: Workato connects seamlessly to Fabric via Azure Blob / OneLake ADLS Gen2 APIs or SQL Endpoints to drop raw Parquet/JSON payloads into Bronze Lakehouses.


MACH Systems
Salesforce CRM
CommerceLayer OMS
Sage Intacct Accounting
PIMcore GCP
Swoogo
Brightspace
Stripe
Webhooks / Bulk APIs
Workato iPaaS
Extract & Load Recipes
Append Raw Files
Microsoft Fabric OneLake



6. Implementation Roadmap

1. **Phase 1: Workato Landing Setup**
   - Configure Workato environment connection to Microsoft Fabric OneLake storage (ADLS Gen2 / REST endpoint).

- Build Bronze layer ingestion recipes for all key sources: Salesforce, CommerceLayer, Sage Intacct, Stripe, Swoogo, Brightspace, and PIMcore.

2. **Phase 2: Data Lakehouse Architecture in Fabric**
   - Provision Microsoft Fabric Capacity (F-SKU).
   - Establish Bronze, Silver, and Gold Lakehouse workspaces.
   - Implement PySpark / T-SQL transformation pipelines for data cleansing and dimensional modeling (Member, Order, Financial GL, Event, Course dimensions & fact tables).
3. **Phase 3: Power BI Direct Lake Semantic Models**
   - Build Power BI semantic models over Gold Delta tables using Direct Lake mode.
   - Design enterprise dashboards for Member 360, E-Commerce Order Performance, Financial Ledger Reconciliation, Event Performance, and CPD Learning Analytics.
4. **Phase 4: Governance & Security**
   - Configure Entra ID role-based access control (RBAC) and Row-Level Security (RLS) across Fabric and Power BI.